1 May 1995

PHYSICS LETTERS A

ELSEVIER

Physics Letters A 200 (1995) 340–344

# Violating Bell inequality by mixed spin-$\frac{1}{2}$ states: necessary and sufficient condition

R. Horodecki [a,1], P. Horodecki [b], M. Horodecki [b]

[a] *Institute of Theoretical Physics and Astrophysics, University of Gdańsk, 80-952 Gdańsk, Poland*

[b] *Departament of Mathematics and Physics, University of Gdańsk, 80-952 Gdańsk, Poland*

Received 4 January 1995; accepted for publication 27 February 1995

Communicated by V.M. Agranovich

**Abstract**

The necessary and sufficient condition for violating the Clauser–Horne–Shimony–Holt (CHSH) inequality by an arbitrary mixed spin-$\frac{1}{2}$ state is presented. Some examples of mixtures which demonstrate the utility of the condition are considered. In particular, it is shown that the local hidden variable (LHV) model for mixed states introduced by Werner [Phys. Rev. A 40 (1989) 4277] is forbidden in some region.

The well known contradiction between quantum theory and local realism was originally proved [1–3] by means of Bell type inequalities relating averages of four random dichotomic variables $a$, $a'$ and $b$, $b'$ representing measurements of spin in directions $\hat{a}$, $\hat{a}'$, $\hat{b}$, $\hat{b}'$ respectively. The problem of violating the Bell type inequalities has been completely solved [4–7] for pure states. However, "for mixtures not even the simplest case (two-spin-$\frac{1}{2}$) is solved" [8]. A natural conjecture was that only mixed states that are convex combinations of product states admit the local hidden variable (LHV) model. This conjecture was contradicted by Werner [9], who gave a counterexample. He started from a family of mixed states (called here Werner mixtures) and showed that some of them admit the LHV model. A special case (spin-$\frac{1}{2}$) was considered by Popescu [8] in the context of quantum teleportation. Some other results concerning the violating Bell inequalities by mixed states have been presented. Andås [10] showed that a convex combination of some product spin-$\frac{1}{2}$ states does not violate Bell inequalities for the generalized Bell type observable. Braunstein, Mann, and Revzen showed [11] that some mixed states can produce maximal violations in the Bell inequality due to Clauser, Horne, Shimony and Holt (CHSH) [2]. In this Letter we present an effective criterion for violating the CHSH inequality by an arbitrary mixed spin-$\frac{1}{2}$ state and we show how it works in some examples. In particular we will show that the LHV model for the Werner mixtures is forbidden in some region.

Consider the Hilbert space $\mathcal{H} = C^2 \otimes C^2$. The state on $\mathcal{H}$ can be represented in a Hilbert–Schmidt basis as follows

[1] E-mail: fizrh@halina.univ.gda.pl.



*SSDI* 0375-9601(95)00214-6

$$\varrho = \tfrac{1}{4}\left(I \otimes I + \boldsymbol{r} \cdot \boldsymbol{\sigma} \otimes I + I \otimes \boldsymbol{s} \cdot \boldsymbol{\sigma} + \sum_{n,m=1}^{3} t_{nm}\sigma_n \otimes \sigma_m\right). \tag{1}$$

Here $I$ stands for identity operator, $\{\sigma_n\}_{n=1}^{3}$ are the standard Pauli matrices, $\boldsymbol{r}, \boldsymbol{s}$ are vectors in $\mathbb{R}^3$, $\boldsymbol{r} \cdot \boldsymbol{\sigma} = \sum_{i=1}^{3} r_i\sigma_i$. The coefficients $t_{nm} = \mathrm{Tr}(\varrho\sigma_n \otimes \sigma_m)$ form a real matrix which we shall denote by $T_\varrho$. The condition $\mathrm{Tr}(\varrho^2) \leqslant 1$ implies

$$\sum_{i=1}^{3}(r_i^2 + s_i^2) + \sum_{n,m=1}^{3} t_{nm}^2 \leqslant 3 \tag{2}$$

where the equality is achieved for the pure states. Note that (1) and (2) do not include the nonnegativity condition, which must be assumed separately.

One can easily find that the Bell operator associated with the quantum CHSH inequality has the following general [2] form in our case,

$$\mathcal{B}_{\mathrm{CHSH}} = \hat{\boldsymbol{a}} \cdot \boldsymbol{\sigma} \otimes (\hat{\boldsymbol{b}} + \hat{\boldsymbol{b}}') \cdot \boldsymbol{\sigma} + \hat{\boldsymbol{a}}' \cdot \boldsymbol{\sigma} \otimes (\hat{\boldsymbol{b}} - \hat{\boldsymbol{b}}') \cdot \boldsymbol{\sigma}, \tag{3}$$

where $\hat{\boldsymbol{a}}, \hat{\boldsymbol{a}}', \hat{\boldsymbol{b}}, \hat{\boldsymbol{b}}'$ are unit vectors in $\mathrm{R}^3$. Then the CHSH inequality is

$$|\langle\mathcal{B}_{\mathrm{CHSH}}\rangle_\varrho| \leqslant 2, \tag{4}$$

where $\langle\mathcal{B}_{\mathrm{CHSH}}\rangle_\varrho \equiv \mathrm{Tr}(\varrho\mathcal{B}_{\mathrm{CHSH}})$. Before we formulate the theorem, let us introduce some quantities. The matrix $U_\varrho := T_\varrho^{\mathrm{T}}T_\varrho$ is a symmetric one, so it can be diagonalized ($T_\varrho^{\mathrm{T}}$ stands for transposition of $T_\varrho$). We denote two greater, obviously positive, eigenvalues of $U_\varrho$ by $u, \tilde{u}$ and by $\|\ \ \|$ the Euclidean norm in $\mathbb{R}^3$. Then we define a quantity

$$M(\varrho) := \max_{\hat{c},\hat{c}'}(\|T_\varrho\hat{\boldsymbol{c}}\|^2 + \|T_\varrho\hat{\boldsymbol{c}}'\|^2) = u + \tilde{u}, \tag{5}$$

where the maximum is taken over all unit and mutually orthogonal pairs of vectors in $\mathbb{R}^3$. This equality can be easily found by putting $\|T_\varrho\hat{\boldsymbol{c}}\|^2 = (\hat{\boldsymbol{c}}, T_\varrho^{\mathrm{T}}T_\varrho\hat{\boldsymbol{c}})$ and using the Lagrange multipliers method (the brackets stand for Euclidean scalar product in $\mathbb{R}^3$). Now we can formulate the main result of this paper.

*Theorem.* The density matrix (1) violates inequality (4) for some operator of the form (3) iff $M(\varrho) > 1$.

To prove the theorem we shall compute the mean value of an arbitrary $\mathcal{B}_{\mathrm{CHSH}}$ in an arbitrary fixed state $\varrho$ and then maximize it with respect to all $\mathcal{B}_{\mathrm{CHSH}}$. A simple calculation gives us

$$\langle\mathcal{B}_{\mathrm{CHSH}}\rangle_\varrho = (\hat{\boldsymbol{a}}, T_\varrho(\hat{\boldsymbol{b}} + \hat{\boldsymbol{b}}')) + (\hat{\boldsymbol{a}}', T_\varrho(\hat{\boldsymbol{b}} - \hat{\boldsymbol{b}}')). \tag{6}$$

Following Ref. [7] we introduce a pair of unit and mutually orthogonal vectors $\hat{\boldsymbol{c}}, \hat{\boldsymbol{c}}'$ by

$$\hat{\boldsymbol{b}} + \hat{\boldsymbol{b}}' = 2\cos\theta\,\hat{\boldsymbol{c}}, \qquad \hat{\boldsymbol{b}} - \hat{\boldsymbol{b}}' = 2\sin\theta\,\hat{\boldsymbol{c}}' \tag{7}$$

where $\theta \in [0, \tfrac{1}{2}\pi]$. Then we find the maximum of (6) as follows,

[2] One can omit the identity, which is formally a dichotomic observable as it commutes with each operator and so its presence can produce no violation (see Ref. [11])

$$\max_{\mathcal{B}_{\text{CHSH}}} \langle \mathcal{B}_{\text{CHSH}} \rangle_\varrho = \max_{\theta,\hat{a},\hat{a}',\hat{c},\hat{c}'} 2[(\hat{a}, T_\varrho \hat{c}) \cos\theta + (\hat{a}', T_\varrho \hat{c}') \sin\theta]$$

$$= \max_{\theta,\hat{c},\hat{c}'} 2[\,\|T_\varrho \hat{c}\| \cos\theta + \|T_\varrho \hat{c}'\| \sin\theta] = \max_{\hat{c},\hat{c}'} 2\sqrt{\|T_\varrho \hat{c}\|^2 + \|T_\varrho \hat{c}'\|^2} = 2\sqrt{M(\varrho)}. \tag{8}$$

Now one can take in turn: $\hat{c}_{\max}$, and $\hat{c}'_{\max}$ chosen as the $U_\varrho$ eigenvectors maximalizing $M(\varrho)$; $\hat{a}_{\max}$, and $\hat{a}'_{\max}$ chosen as unit vectors in the directions $T_\varrho \hat{c}_{\max}$, $T_\varrho \hat{c}'_{\max}$; $\theta_{\max}$ defined by $\|T_\varrho \hat{c}_{\max}\| \sin\theta_{\max} = \|T_\varrho \hat{c}'_{\max}\| \cos\theta_{\max}$. Having all these (generally nonunique) parameters one can construct the observable $\mathcal{B}_{\max}$. It provides both the maximal and the minimal value of (6). The latter one is achieved by $-\mathcal{B}_{\max}$. Under this symmetry we get

$$2\sqrt{M(\varrho)} = \langle \mathcal{B}_{\max} \rangle_\varrho = \max_{\mathcal{B}_{\text{CHSH}}} |\langle \mathcal{B}_{\text{CHSH}} \rangle_\varrho| \tag{9}$$

so the inequality (4) is violated iff $\langle \mathcal{B}_{\max} \rangle_\varrho = 2\sqrt{M(\varrho)} > 2$.

Note that since $|\langle \mathcal{B}_{\text{CHSH}} \rangle_\varrho| \leqslant 2\sqrt{2}$ [3] the inequality $M(\varrho) \leqslant 2$ must hold. The following estimate,

$$M(\varrho) \leqslant \text{Tr}(U_\varrho) = \sum_{i,j=1}^{3} t_{ij}^2 \leqslant 3 - \sum_{i=1}^{3} (r_i^2 + s_i^2), \tag{10}$$

obtained from (2), (5) may be useful in some cases. Some remarks to find $M(\varrho)$ for particular cases should be made: (i) If $T_\varrho$ is either an orthogonal matrix or a singular one acting orthogonally in the two dimensions then $M(\varrho) = 2$, i.e. $\varrho$ violates the CHSH inequality maximally. (ii) If $T_\varrho$ is symmetric, then one can diagonalize it and $M(\varrho) = t^2 + \tilde{t}^2$, where $t$, and $\tilde{t}$ are the $T_\varrho$ eigenvalues having two greater absolute values. Moreover, if $T_\varrho$ is symmetric and singular, then $M(\varrho) = \sum_{i,j=1}^{3} t_{ij}^2$. Indeed $t^2 + \tilde{t}^2 = \text{Tr}(U_\varrho) = \sum_{i,j=1}^{3} t_{ij}^2$, where the invariance of the trace under unitary transformations is used.

Now, we shall give the general form of $T_\varrho$ with respect to the spectral representation of $\varrho$. Namely, every state $\varrho$ in the spin-$\frac{1}{2}$ case can be represented in the following form,

$$\varrho = \sum_{i=1}^{4} p_i |\psi_i\rangle\langle\psi_i|, \qquad \text{where } \langle\psi_i|\psi_j\rangle = \delta_{ij}. \tag{11}$$

The representation is unique for the nondegenerate spectra and every state $|\psi_i\rangle$ has the form $|\psi_i\rangle = a_i e_1 \otimes e_1 + b_i e_1 \otimes e_2 + c_i e_2 \otimes e_1 + d_i e_2 \otimes e_2$ with $|a_i|^2 + |b_i|^2 + |c_i|^2 + |d_i|^2 = 1$. Then

$$T_\varrho = \sum_{i=1}^{4} p_i T_i, \tag{12}$$

where

$$T_i = 2 \begin{pmatrix} \text{Re}(a_i \overline{d}_i + b_i \overline{c}_i) & \text{Im}(a_i \overline{d}_i - b_i \overline{c}_i) & \text{Re}(a_i \overline{c}_i - b_i \overline{d}_i) \\ \text{Im}(a_i \overline{d}_i + b_i \overline{c}_i) & \text{Re}(-a_i \overline{d}_i + b_i \overline{c}_i) & \text{Im}(a_i \overline{c}_i - b_i \overline{d}_i) \\ \text{Re}(a_i \overline{b}_i - c_i \overline{d}_i) & \text{Im}(a_i \overline{b}_i - c_i \overline{d}_i) & \frac{1}{2}(|a_i|^2 - |b_i|^2 - |c_i|^2 + |d_i|^2) \end{pmatrix}. \tag{13}$$

From the above form of $T_\varrho$ follows $|t_{nm}| \leqslant 1$, $n, m = 1, 2, 3$ for every state $\varrho$.

*Examples.* Consider an arbitrary convex combination of product spin-$\frac{1}{2}$ states

$$\varrho = \sum_{i=1}^{N} p_i \varrho_i \otimes \tilde{\varrho}_i, \tag{14}$$

where $\varrho_i = \frac{1}{2}(I + \boldsymbol{r}^i\sigma)$, $\tilde{\varrho}_i = \frac{1}{2}(I + \boldsymbol{s}^i\sigma)$; $\|\boldsymbol{s}^i\|^2, \|\boldsymbol{r}^i\|^2 \leqslant 1$, $i = 1,\dots,N$ as $\mathrm{Tr}(\varrho_i^2)$, $\mathrm{Tr}(\tilde{\varrho}_i^2) \leqslant 1$. The states of type (14) are found to admit the LHV model [9] thus they should not violate the CHSH inequality. It is straightforward to find the elements of the matrix $T_\varrho$,

$$t_{nm} = \sum_{i=1}^{N} p_i r^i_m s^i_n, \qquad m, n = 1, 2, 3. \tag{15}$$

For arbitrary mutually orthogonal unit vectors we get

$$\begin{aligned}\|T_\varrho \hat{\boldsymbol{c}}\|^2 + \|T_\varrho \hat{\boldsymbol{c}}'\|^2 &= \sum_{i,j=1}^{N} p_i p_j (\boldsymbol{r}^i, \boldsymbol{r}^j)[(\boldsymbol{s}^i, \hat{\boldsymbol{c}})(\boldsymbol{s}^j, \hat{\boldsymbol{c}}) + (\boldsymbol{s}^i, \hat{\boldsymbol{c}}')(\boldsymbol{s}^j, \hat{\boldsymbol{c}}')]\\
&\leqslant \tfrac{1}{2}\sum_{i,j=1}^{N} p_i p_j (\boldsymbol{r}^i, \boldsymbol{r}^j)[(\boldsymbol{s}^i, \hat{\boldsymbol{c}})^2 + (\boldsymbol{s}^i, \hat{\boldsymbol{c}}')^2 + (\boldsymbol{s}^j, \hat{\boldsymbol{c}})^2 + (\boldsymbol{s}^j, \hat{\boldsymbol{c}}')^2]\\
&\leqslant \tfrac{1}{2}\sum_{i,j=1}^{N} p_i p_j (\boldsymbol{r}^i, \boldsymbol{r}^j)\left[\|\boldsymbol{s}^i\|^2 + \|\boldsymbol{s}^j\|^2\right] \leqslant \left\|\sum_{i=1}^{N} p_i \boldsymbol{r}^i\right\|^2 \leqslant \left(\sum_{i=1}^{N} p_i \|\boldsymbol{r}^i\|\right)^2 \leqslant 1.\end{aligned} \tag{16}$$

Taking the maximum of the left hand side of the inequality we get the expected estimate.

Now consider the states introduced by Werner [9],

$$\varrho = (d^3 - d)^{-1}[(d - \phi)I + (d\phi - 1)V], \tag{17}$$

where $\mathcal{H} = C^d \otimes C^d$, $d \geqslant 2$; $-1 \leqslant \phi < 0$ and $V$ is defined as $V\varphi \otimes \tilde{\varphi} = \tilde{\varphi} \otimes \varphi$. Those states are not of the form (14) but they commute with each $U \otimes U$ ($U$ denotes a unitary operator on $C^d$). Werner has proved the existence of the LHV model for

$$\phi = -1 + d^{-2}(d + 1). \tag{18}$$

Now, as the CHSH inequality is a necessary condition for the LHV model our criterion $M(\varrho) = \frac{2}{9}(2\phi - 1)^2 > 1$ excludes its existence for $\phi \in [-1, -\frac{3}{4}\sqrt{2} + \frac{1}{2})$ and $d = 2$. Of course $\phi = -\frac{1}{4}$ given by (18) with $d = 2$ is beyond the forbidden interval.

As the last example, we consider a mixture of the pure entangled states,

$$\varrho = \sum_{i=1}^{4} p_i |\phi_i\rangle\langle\phi_i|, \tag{19}$$

where

$$\phi_{\substack{1\\(2)}} = \frac{1}{\sqrt{2}}(e_1 \otimes e_1 \pm e_2 \otimes e_2), \qquad \phi_{\substack{3\\(4)}} = \frac{1}{\sqrt{2}}(e_1 \otimes e_2 \pm e_2 \otimes e_1). \tag{20}$$

Using the formulas (12), (13) we obtain

$$T_\varrho = \mathrm{diag}[2(p_1 + p_3) - 1, 2(p_2 + p_3) - 1, 2(p_1 + p_2) - 1]. \tag{21}$$

As it was mentioned above we have $|t_{ii}| \leqslant 1$ and, apart from the pure state case, neither of the two eigenvalues can simultaneously reach the unit absolute value. Hence no mixture of the above form can violate the CHSH inequality maximally. Obviously this fact does not contradict the results of Ref. [11] as the authors obtained the maximal violation by mixed states for a Hilbert space of a larger dimension (at least $\mathcal{H} = C^4 \otimes C^4$). It is easy to see from (21) that, except for a convex combination of product states ($p_i = p_j = \frac{1}{2}, i \neq j; p_k = 0, k \neq i, j$), each mixture of two states of the form (20) violates the inequality.

In conclusion, our theorem provides an effective criterion for violating the Bell type inequalities by mixed spin-$\frac{1}{2}$ states. So far, given a mixed state it was hard to say whether it violates the CHSH inequality because one had to construct a respective Bell operator for it. On the other hand, given a mixed state there was no way to ensure whether it satisfies the CHSH inequality for each Bell observable (3). The above theorem solves this problem completely in terms of the parameters of the density operator itself. Using it we have got, in particular, the results which are compatible with Ref. [9]. Moreover, we have shown that the criterion excludes the LHV model for Werner mixtures with $\phi \in [-1, -\frac{3}{4}\sqrt{2} + \frac{1}{2})$, $d = 2$. Finally we believe that the present result will be a useful tool in further investigations of the aspects of the quantum nonseparability discussed recently in the context of the information-theoretic approach to correlated quantum systems [12] and the one involving the nonstandard local measurements performed on the two separated systems [8,13]. In particular, it has been suggested that such measurements may reveal a "hidden nonlocality" despite the existence of the LHV model [13]. Surprisingly, apart from the nonstandard measurement concept, the CHSH inequality was used in Ref. [13].

We would like to thank Robert Alicki and Marek Żukowski for useful comments and discussions. We are indebted to Sandu Popescu for sending the preprints of his papers.

## References

[1] S. Bell, Physics 1 (1964) 195.
[2] J.F. Clauser, M.A. Horne, A. Shimony and R.A. Holt, Phys. Rev. Lett. 23 (1969) 880.
[3] B.S. Cirel'son, Lett. Math. Phys. 4 (1980) 93.
[4] V. Capasso, D. Fortunato and F. Selleri, Int. J. Mod. Phys. 7 (1973) 319.
[5] N. Gissin, Phys. Lett. A 154 (1991) 201.
[6] N. Gissin and A. Peres, Phys. Lett. A 162 (1992) 15.
[7] S. Popescu and D. Rohrlich, Phys. Lett. A 166 (1992) 293.
[8] S. Popescu, Phys. Rev. Lett. 72 (1994) 779.
[9] R.F. Werner, Phys. Rev. A 40 (1989) 4277.
[10] H.E. Andås, Phys. Lett. A 167 (1992) 6.
[11] S.L. Braunstein, A. Mann, M. Revzen, Phys. Rev. Lett. 68 (1992) 3259.
[12] R. Horodecki and P. Horodecki, Phys. Lett. A 194 (1994) 147.
[13] S. Popescu, Technion – Israel Institute of Technology preprint TECH-PH-94-1.